

Problem Set 16 Discussion

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- tr is unhappy with multi-byte Unicode characters

Produces three “EEE” in the output

```
tr '$'\u2020' "E" < IOC_14.2.csv >IOC_14.2_new.csv
```

- Option 1 – squeeze repeats
 - The tr -s flag is the “squeeze repeats” flag for the rightmost set (in this case EEE > E)

Adding the -s flag

```
tr -s '$'\u2020' 'E' < IOC_14.2.csv >IOC_14.2_new.csv
```

- Option 2 – convert hexadecimal to octal and use backslashes
 - The typical Unicode characters are encoded with hexadecimal
 - You can use octal (UTF-8) encoding of the same information

Converting hexadecimal to octal

```
tr '\xE2\x80\xA0' 'E' < IOC_14.2-2.csv > IOC_14.2-2_new.csv
```